

Copula Entropy

Theory and Applications

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- Lecture 6: Evaluation
- Lecture 7: Generalizations
- Lecture 8: Real applications and softwares

Lecture 3: Theoretical applications (1)

① Association Discovery

② Structure Learning

③ Variable Selection

Copula Entropy: Application 1

Association Discovery¹

¹Jian Ma. "Discovering Association with Copula Entropy". In: *arXiv preprint arXiv:1907.12268* (2019).

Copula Entropy: Association Discovery

- Problem
 - To discover association relations between random variables from data
- History
 - An old and fundamental problem date back to the early days of statistics
- Related Methods
 - Pearson Correlation Coefficient r
 - Spearman's ρ and Kendall's τ
 - Regression

Copula Entropy: Association Discovery

Traditional association measures

- Pearson Correlation Coefficient²

$$r_{XY} = \text{corr}(X, Y) = \frac{\text{cov}(X, Y)}{\delta_X \delta_Y} \quad (1)$$

²Karl Pearson. "Mathematical contributions to the theory of evolution.—On a form of spurious correlation which may arise when indices are used in the measurement of organs". In: *Proceedings of The Royal Society of London* 60.1 (1896), pp. 489–498.

Copula Entropy: Association Discovery

Traditional association measures

- Spearman's ρ^3

$$\rho_{XY} = \frac{1 - 6 \sum_i (r_i - s_i)^2}{n(n^2 - 1)} \quad (2)$$

$$= 12 \int_u \int_v C(u, v) du dv - 3 \quad (3)$$

³C. Spearman. "The Proof and Measurement of Association between Two Things". In: *The American Journal of Psychology* 100.3/4 (1987), pp. 441–471.

Copula Entropy: Association Discovery

Traditional association measures

- Kendall's τ^4

$$\tau_{XY} = 1 - \frac{4Q}{n(n-1)} \quad (4)$$

$$= 4 \int_u \int_v C(u, v) dC(u, v) - 1 \quad (5)$$

⁴M. G. Kendall. "A New Measure of Rank Correlation". In: *Biometrika* 30.1/2 (1938), pp. 81–93. ISSN: 00063444. URL: <http://www.jstor.org/stable/2332226>.

Copula Entropy: Association Discovery

Traditional association measures

- Gini's γ^5

$$\gamma_{X,Y} = \frac{1}{D_n} \sum_{i=1}^n (|n+1 - r_i - s_i| - |r_i - s_i|) \quad (6)$$

$$= 2 \int_u \int_v (|u+v-1| - |u-v|) dC(u, v). \quad (7)$$

⁵Corrado Gini. *L'ammontare e la composizione della ricchezza delle nazioni*. Vol. 62. Fratelli Bocca, 1914.

Copula Entropy: Association Discovery

Traditional association measures

- Schweizer and Wolff's σ^6

$$\sigma_{X,Y} = 12 \int_u \int_v |C(u, v) - uv| du dv. \quad (8)$$

⁶B. Schweizer and E. F. Wolff. "On Nonparametric Measures of Dependence for Random Variables". In: *The Annals of Statistics* 9.4 (1981), pp. 879–885. DOI: 10.1214/aos/1176345528. URL: <https://doi.org/10.1214/aos/1176345528>.

Copula Entropy: Association Discovery

Why Copula Entropy?

Table: Theoretical comparison between CE and CC.

	CC	CE
linearity	linear	nonlinear
Order	2	≥ 2
Assumption	Gaussianity	None
variate	bivariate	multivariate

Copula Entropy: Association Discovery

CE between random vectors

- Theorem

$$H_c(\mathbf{x}_1; \dots; \mathbf{x}_n) = H_c(\mathbf{x}) - \sum_{i=1}^n H_c(\mathbf{x}_i). \quad (9)$$

- Merits
 - rigorous and provable
 - intuitive and natural
 - easy to implement

Copula Entropy: Association Discovery

CE between random vectors

Proof.

$$H_c(\mathbf{x}_1; \dots; \mathbf{x}_n) = -I(\mathbf{x}_1; \dots; \mathbf{x}_n) \quad (10)$$

$$= - \int_{\mathbf{x}} p(\mathbf{x}) \log \frac{p(\mathbf{x})}{p(\mathbf{x}_1) \cdots p(\mathbf{x}_n)} d\mathbf{x} \quad (11)$$

$$= -I(\mathbf{x}) + \sum_{i=1}^n I(\mathbf{x}_i) \quad (12)$$

$$= H_c(\mathbf{x}) - \sum_{i=1}^n H_c(\mathbf{x}_i). \quad (13)$$

Here if $d_i = 1$, then $H_c(\mathbf{x}_i) = 0$.



Copula Entropy: Association Discovery

Experiments on the NHANES data

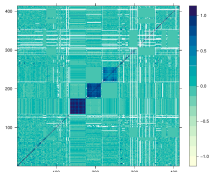
- Objectives of NHANES
 - to monitor trends and emerging issues of population health
 - to investigate its relationship with risk factors, nutrition and environmental exposures, etc.
- NHANES (2013-2014)
 - 14,332 persons from 30 different survey locations were selected
 - Of those selected, 10,175 interviewed and 9,813 examined
 - 5 groups of data: demographics, dietary, examination, laboratory, and questionnaire
- Experimental data

The laboratory data, which includes 423 variables from blood, urine, oral rinse and vaginal/Penile swabs.
- Missing values

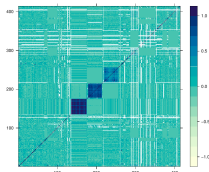
The missing values were filled with the mean of their corresponding variables.

Copula Entropy: Association Discovery

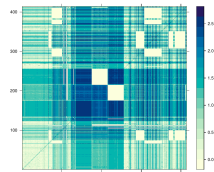
Results - Correlation matrices



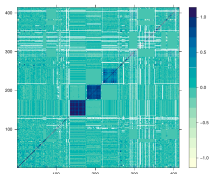
Pearson's r



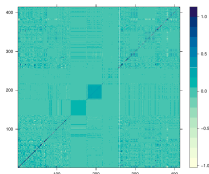
Kendall's τ



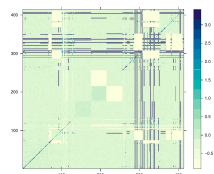
Schweizer & Wolff's σ



Spearman's ρ



Gini's γ



CE

Copula Entropy: Application 2

Structure Learning⁷

⁷Ma2009, Jian Ma and Zengqi Sun. "Dependence Structure Estimation via Copula". In: *arXiv preprint arXiv:0804.4451* (2008).

Copula Entropy: Structure Learning

- Problem
 - To learn statistical structure among random variables from data
- Graph Representation
 - A probability density is represented with a directed or undirected graph, of which each node represents a random variable, and each edge represents a (conditional) dependence relation between two random variables
- Related Methods
 - Chow-Liu Algorithm
 - Vine Copula

Copula Entropy: Structure Learning

Given random variables $\mathbf{X} = (X_1, \dots, X_m)$ and their corresponding graph structure G , then their joint density function can be represented as the following product form:

$$p(\mathbf{x}|G) = \prod_{i=1}^m p(x_i|x_{\pi_i}), \quad (14)$$

where π_i is the predecessor nodes of x_i in G .

Graph representation has its corresponding copula, since graph represents dependence relationships and so does copula. In this sense, (14) can be transformed into the following copula form:

$$p(\mathbf{x}|G) = \prod_{i=1}^m c_i(x_i, x_{\pi_i}) \prod_{i=1}^m p(x_i). \quad (15)$$

Copula Entropy: Structure Learning

Chow-Liu method⁸ generates optimal tree structure to approximate density function by maximizing the following MI sum of tree structure with minimal-spanning-tree (MST) algorithm:

$$\max \sum_{i=1}^m I(x_i, x_{\pi_i}). \quad (16)$$

⁸C.K. Chow and C.N. Liu. "Approximating discrete probability distributions with dependence trees". In: *IEEE Transactions on Information Theory* 14.3 (1968), pp. 462–467.

Copula Entropy: Structure Learning

Given a sample $\mathbf{X} = X_1, \dots, X_T$ from $p(\mathbf{x})$ associated with copula function $c(\mathbf{u})$, then

- Likelihood

$$\mathcal{L}(\theta; \mathbf{X}) = \sum_{t=1}^T \log p(\mathbf{x}_t; \theta) \quad (17)$$

- **Copula Likelihood**⁹

$$\mathcal{L}_c(\theta_c; \mathbf{X}) = \sum_{t=1}^T \log c(\mathbf{u}_t; \theta_c). \quad (18)$$

- equivalent to copula entropy
- guide principle of copula based structure learning

⁹ 马健. “Copula 结构学习”. 博士学位论文. 清华大学, 2009.

Copula Entropy: Structure Learning

- **Copula Entropy based Chow-Liu Algorithm**

- ① computing dependence matrix $\mathbf{W}_x$ of data x with CE estimation
- ② constructing dependence structure T from $\mathbf{W}_x$ with MST algorithm

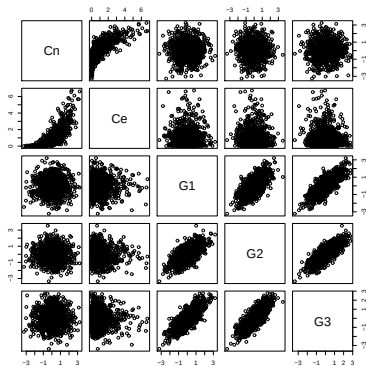
- **Advantages**

- distribution-free, non-parametric
- tuning-free, insensitive to parameters
- easy to implement
- low computation burden

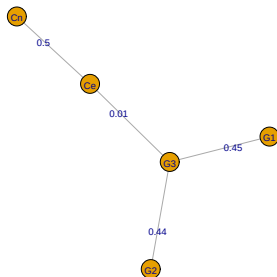
Copula Entropy: Structure Learning

Simulation Experiment

- 5 random variables: the first three are Gaussian and the others two are governed by Gaussian copula with margins as normal distribution and exponential distribution respectively



Simulated data



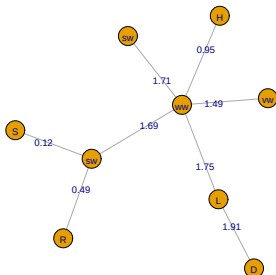
Learned graph

Copula Entropy: Structure Learning

Experiment on real data

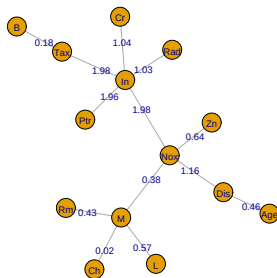
- Abalone data

Predicting the age of abalone from physical measurements



- Boston housing data

Concerns housing values in suburbs of Boston



Copula Entropy: Application 3

Variable Selection¹⁰

¹⁰ Jian Ma. "Variable Selection with Copula Entropy". In: *Chinese Journal of Applied Probability and Statistics* 37.4 (2021). See also arXiv preprint [arXiv:1910.12389](https://arxiv.org/abs/1910.12389), 2019, pp. 405–420.

Copula Entropy: Variable Selection

- Problem
 - To select a 'right' subset of variables from the whole group for building classification or regression models with good predictability and interpretability
- History
 - An old and basic problem in statistics and machine learning
- Related Problems
 - Feature Selection
 - Model Selection

Copula Entropy: Variable Selection

Existing methods - Information Criteria

likelihood with penalty on the number of parameters in the models

- AIC¹¹

$$\text{AIC} = -2L + 2p \quad (19)$$

- BIC¹²

$$\text{BIC} = -2L + p \log N \quad (20)$$

¹¹Hirotsugu Akaike. "A new look at the statistical model identification". In: *IEEE Transactions on Automatic Control* 19.6 (1974), pp. 716–723, Hirotsugu Akaike. "Information Theory and an Extension of the Maximum Likelihood Principle". In: *Selected Papers of Hirotsugu Akaike*. Ed. by Emanuel Parzen, Kunio Tanabe, and Genshiro Kitagawa. Springer New York, NY, 1998, pp. 199–213. DOI: 10.1007/978-1-4612-1694-0_15. URL: https://doi.org/10.1007/978-1-4612-1694-0_15.

¹²Gideon Schwarz. "Estimating the Dimension of a Model". In: *The Annals of Statistics* 6.2 (1978), pp. 461–464.

Copula Entropy: Variable Selection

Existing methods - Penalized GLMs

likelihood with penalty on the nonzero coefficients in the GLMs

- LASSO¹³
- Ridge Regression¹⁴
- Elastic Net¹⁵

$$\min_{\beta} \{L(\beta; y, \mathbf{X}) + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2^2\} \quad (21)$$

- Adaptive LASSO¹⁶

$$\min_{\beta} \{L(\beta; y, \mathbf{X}) + \lambda \sum_{j=1}^p w_j |\beta_j|\} \quad (22)$$

¹³Robert Tibshirani. "Regression Shrinkage and Selection via the LASSO". In: *Journal of the Royal Statistical Society Series B-Methodological* 58.1 (1996), pp. 267–288.

¹⁴Arthur E. Hoerl and Robert W. Kennard. "Ridge regression: biased estimation for nonorthogonal problems". In: *Technometrics* 42.1 (2000), pp. 80–86.

¹⁵Hui Zou and Trevor Hastie. "Regularization and variable selection via the elastic net". In: *Journal of The Royal Statistical Society Series B-statistical Methodology* 67.2 (2005), pp. 301–320.

¹⁶Hui Zou. "The adaptive LASSO and its oracle properties". In: *Journal of the American Statistical Association* 101.476 (2006), pp. 1418–1429.

Copula Entropy: Variable Selection

Existing methods - Statistical independence measures

- Distance Correlation¹⁷

$$\text{dCor}(X, Y) = \frac{\nu^2(X, Y)}{\sqrt{\nu^2(X)\nu^2(Y)}}, \quad (23)$$

where $\nu^2(X, Y)$ be distance covariance.

¹⁷Gábor J. Székely, Maria L. Rizzo, and Nail K. Bakirov. "Measuring and testing dependence by correlation of distances". In: *The Annals of Statistics* 35.6 (2007), pp. 2769–2794, Gábor J. Székely and Maria L. Rizzo. "Brownian distance covariance". In: *The Annals of Applied Statistics* 3.4 (2009), pp. 1236–1265.

Copula Entropy: Variable Selection

Existing methods - Statistical independence measures

- Hilbert-Schmidt Independence Criterion (HSIC)¹⁸

$$\mathbf{dHSIC}(P(\mathbf{X})) = \|\Pi(P(X_1) \otimes, \dots, \otimes P(X_d)) - \Pi(P(\mathbf{X}))\|, \quad (24)$$

where Π be the mean embedding function associated with kernel functions.

¹⁸ Arthur Gretton et al. "A Kernel Statistical Test of Independence". In: *Advances in Neural Information Processing Systems* 20. Vol. 20. 2007, pp. 585–592, Niklas Pfister et al. "Kernel-based Tests for Joint Independence". In: *Journal of The Royal Statistical Society Series B-statistical Methodology* 80.1 (2018), pp. 5–31.

Copula Entropy: Variable Selection

- CE based method

To select variables X_i according to the rank of their negative CE values with target Y

$$-H_c(X_i, Y) > -H_c(X_j, Y) > \dots > -H_c(X_k, Y) \quad (25)$$

- Advantages

- model-free, non-parametric
- tuning-free, insensitive to parameters
- interpretable with physical meanings
- supported by rigorous math
- science instead of art, compared with existing methods
- easy to implement, low computation burden

Copula Entropy: Variable Selection

Experiments on the UCI heart disease data¹⁹

- Overview of the data

The data set contains 4 databases (899 samples) concerning heart disease diagnosis. All attributes are numeric-valued. The data was collected from the four following locations:

- Cleveland clinic foundation;
- Hungarian Institute of Cardiology, Budapest;
- V.A. medical center, long beach, CA;
- University hospital, Zurich, Switzerland.

- Attributes

The data has 76 attributes (#58 'num' for diagnosis). Of them, 13 attributes are recommended by professionals as clinical relevant.

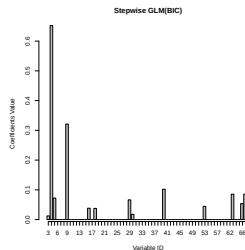
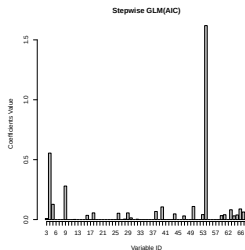
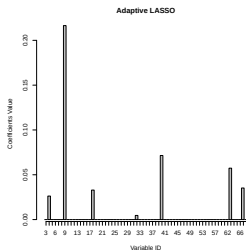
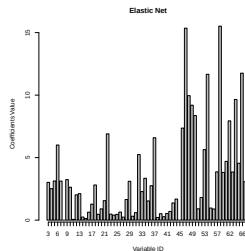
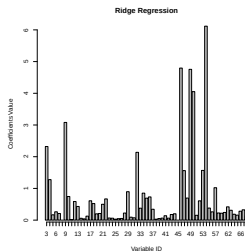
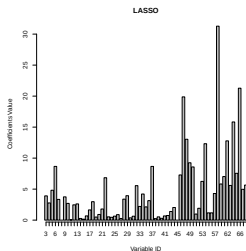
Table: Recommended attributes.

ID	3	4	9	10	12	16	19
Name	age	sex	cp	trestbps	chol	fbs	restecg
ID	32	38	40	41	44	51	58
Name	thalach	exang	oldpeak	slope	ca	thal	num

¹⁹ Asuncion2007.

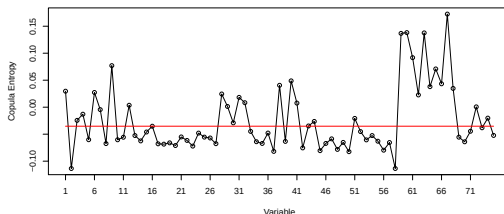
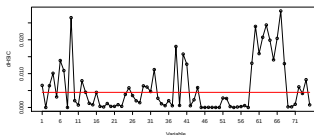
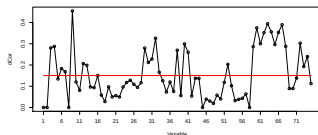
Copula Entropy: Variable Selection

- Results** - Coefficients of penalized likelihood based models



Copula Entropy: Variable Selection

- **Results** - with statistical dependence measures (dCor, dHSIC, CE)



Copula Entropy: Variable Selection

- Results** - Prediction accuracy

the selected variables present the best prediction accuracy.

Model	Accuracy(%)
SVM(Recommended variables)	84.20
SVM(CE)	84.76
SVM(dCor)	82.76
SVM(dHSIC)	84.54
Stepwise GLM(AIC)	51.8
Stepwise GLM(BIC)	49.1
LASSO	79.2
Ridge Regression	63.0
Elastic Net	75.9
Adaptive LASSO	35.7

Copula Entropy: Variable Selection

- Results** - Selected variables

Copula Entropy selects more 'right' variables than the other methods do.

Method	Selected Variables' ID	✓
Recommended variables	3,4,9,10,12,16,19,32,38,40,41,44,51	13
CE	3,4,6,7,9,12,16,28-32,38,40,41,44,51,59-68	11
dHSIC	3,4,6,7,9,12,13,16,25,29-32,38,40,41,44,59-68	10
dCor	3,4,6,7,9,12,13,16,28-33,38,40,41,52,59-68	9
Stepwise GLM(AIC)	3,4,5,9,12,16,18,20,26,29,30,32,40,44,47,50,53,54,60,61,63,65-67	8
Stepwise GLM(BIC)	3,4,5,9,16,18,29,30,40,53,63,66,67	5
Adaptive LASSO	4,6,9,18,32,40,63,67	4
LASSO		
Ridge Regression	all except 8,45	-
Elastic Net		

Monograph

Jian Ma. “Copula Entropy: Theory and Applications”. In: *arXiv preprint arXiv:2025.18168* (2025)

Thanks!

Q&A